What are your odds of reaching a Himalayan summit?

Predicting climber success with the Himalayan Database

Name 1 Name 2

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Statistical and Machine Learning – WU Vienna

The question

- Himalayan expeditions are expensive, long and dangerous
- Only 1 in 3 climbers reaches the summit
- A climber choosing peak, season and team, or an agency planning a trip, wants to know the odds before leaving

Task: binary classification

$P(\text{summit} \mid \text{what is known before the climb})$

We compare four methods

1. Elastic-net logistic regression
2. Pruned classification tree
3. Random forest
4. Gradient boosting

from easy to interpret $\rightarrow$ most flexible

Data: The Himalayan Database

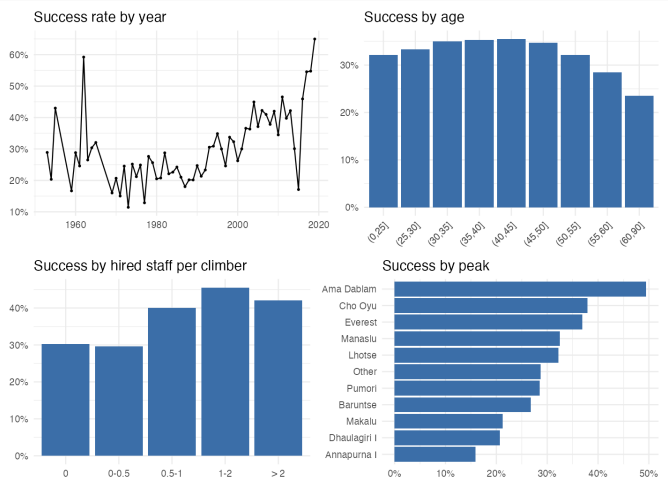
- Records of all expeditions in the Nepal Himalaya (Elizabeth Hawley's archive), via TidyTuesday
- Three tables: **members**, **expeditions**, **peaks**, joined to one row per climber

Population

- Climbers and leaders who actually attempted the climb, 1950–2019
- Hired Sherpa staff excluded: they summit 66% of the time as part of their job

Climbers	Expeditions	Peaks	Summit rate
55,049	9,883	300+	33.8%

Exploration: what goes with success?



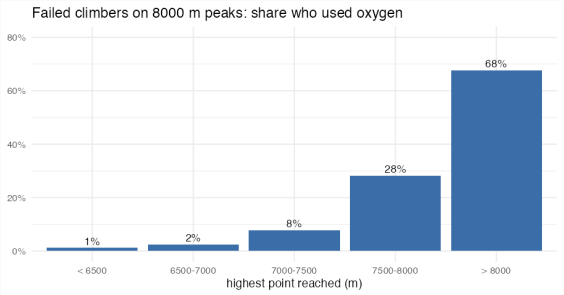
Pre-processing: only use what is known before the climb

Features (16)

- *Climber:* age, sex, citizenship (top 15), leader
- *Mountain:* peak (top 10), height, 8000 m peak, climbed before
- *Expedition:* year, season, team size, hired staff, staff per climber, commercial

Removed (leakage)

highest point reached, death, injury, oxygen use, reason the expedition ended



Oxygen is recorded only if it was used, i.e. once a climber is already high up.

Set-up: split by expedition, judge the probabilities

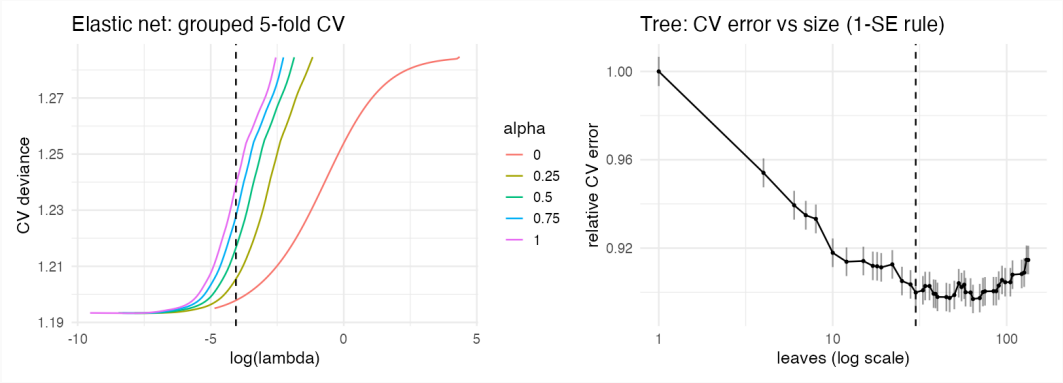
Splitting

- Climbers of one expedition share weather, route and team → never split them
- **Test set:** 20% of expeditions (11,133 climbers), used once at the end
- **Tuning:** 5-fold CV grouped by expedition

Metrics

- **Log loss** (main) and Brier score: quality of the probabilities
- **AUC**: ranking of climbers
- **Calibration**: do 40% predictions come true 40% of the time?
- Accuracy is weak: always saying "no summit" already gets 68%

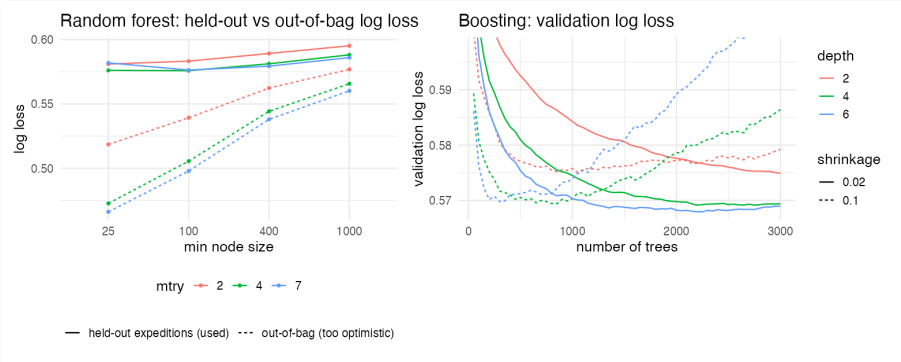
Tuning the interpretable models



Elastic net: spline terms for age, year, height; α and λ by CV, 1-SE rule $\rightarrow$ $\alpha = 0.25$, 38 of 49 terms kept. All α tie, only ridge is slightly worse.

Tree: cost-complexity pruning, 1-SE rule $\rightarrow$ 30 leaves.

Tuning the flexible models: the out-of-bag trap



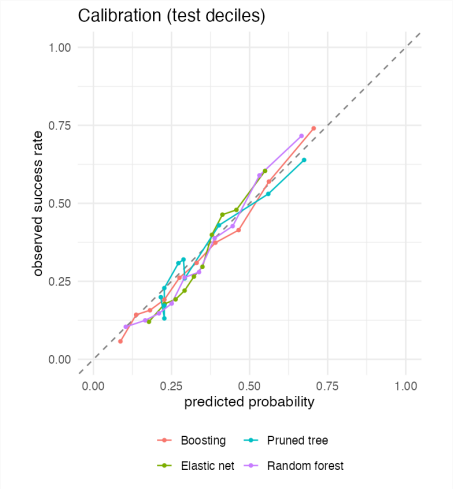
Random forest: out-of-bag error leaves out single *climbers*, so teammates stay in the bag → too optimistic (0.47 vs 0.58). Tuned on held-out expeditions: mtry 4, node size 100.

Boosting: depth × shrinkage × trees on a held-out fold: depth 6, $\nu = 0.02$, 2,250 trees.

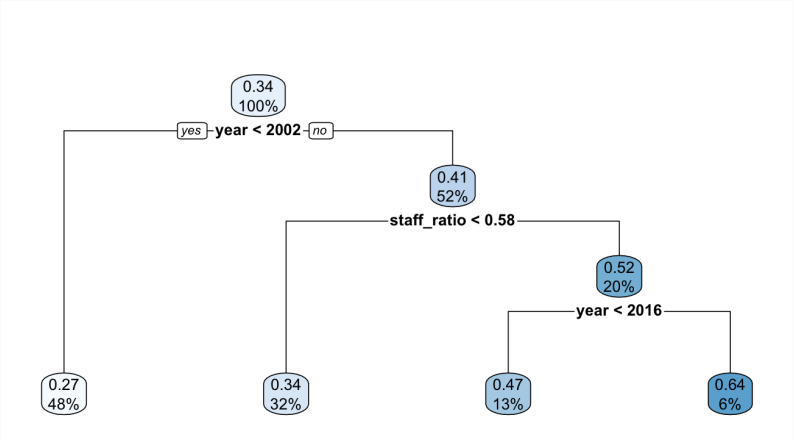
Results on the test set

Model	Log loss	AUC	Acc.
Always “no”	0.629	0.500	0.678
Elastic net	0.583	0.695	0.698
Pruned tree	0.579	0.685	0.716
Random forest	0.541	0.751	0.740
Boosting	**0.534**	**0.756**	0.739

- Flexible models clearly beat the linear model
- Boosting best, forest close
- All models well calibrated

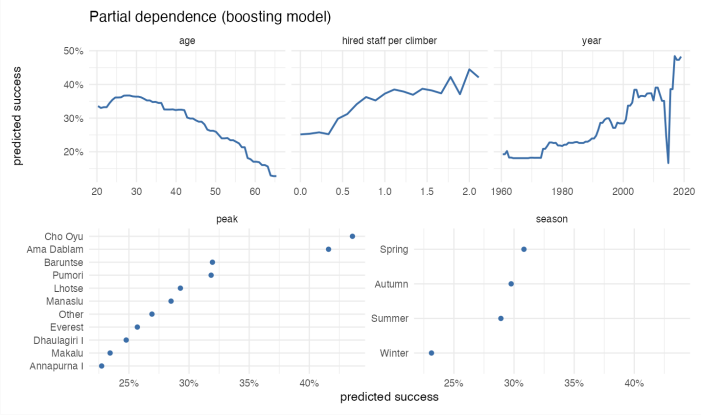


The tree as a rule book



Top 3 splits of the 30-leaf tree (box: success rate, share of climbers). First split is **time** (before 2002: 27%, after: 41%), then **hired staff per climber**: modern teams with ≥ 0.58 staff per climber reach 52%, and 64% since 2016.

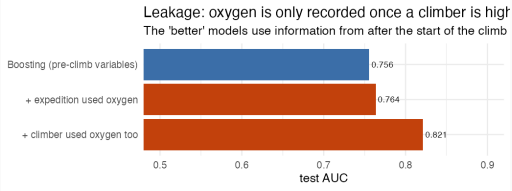
What drives success? (boosting model)



- **Year:** 20% in the 1960s → 48% today; dip in 2014–15 (Everest avalanche, earthquake: 0% on Everest in 2015)
- **Age:** steady decline after 40
- **Sherpa support:** 25% without → ∼40% with 1+ staff per climber
- **Winter** is hardest; Cho Oyu and Ama Dablam the "easiest" peaks

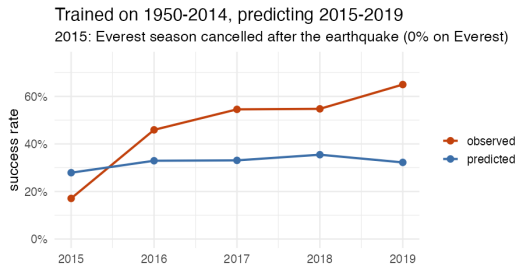
How far can we trust it?

Leakage check



Adding the climber's oxygen use lifts AUC to 0.82, a warning sign, not a better model.

Time split



Trained up to 2014: predicts 33%, reality 47%.

Trees cannot extrapolate the upward trend.

Conclusion

Predicted odds from the boosting model (US climber, 2018)

45-year-old woman, commercial Everest expedition, spring	**67%**
30-year-old man, small team on Ama Dablam, autumn	**55%**
55-year-old man, Everest, no hired staff	**21%**
35-year-old man, Dhaulagiri I, winter	**21%**

Take-aways

1. Boosting is best (AUC 0.76); the gain over the linear model is clear but moderate
2. Odds depend on *when*, *where*, *how much support* and age
3. Grouping by expedition and removing leakage changed the conclusions

Limitations

- No climber experience in the data
- Strong trend over time (see time split)
- Weather and luck are not observed

Thank you!

Questions?

Backup

Backup: full test results

Model	Log loss	Brier	AUC	Accuracy
Always "no summit"	0.629	0.219	0.500	0.678
Elastic net	0.583	0.198	0.695	0.698
Pruned tree	0.579	0.196	0.685	0.716
Random forest	0.541	0.181	0.751	0.740
Boosting	0.534	0.178	0.756	0.739
Leaky: boosting + expedition oxygen	0.527	0.176	0.764	0.739
Leaky: boosting + both oxygen variables	0.470	0.154	0.821	0.780

Test set: 11,133 climbers from 1,977 expeditions. Training: 43,916 climbers.

Backup: tuning grids

Random forest (300 trees, held-out fold)

mtry	node size	held-out / OOB log loss
4	**100**	**0.576** / 0.506
4	25	0.576 / 0.473
7	100	0.576 / 0.498
7	25	0.582 / 0.466
2	1000	0.595 / 0.577

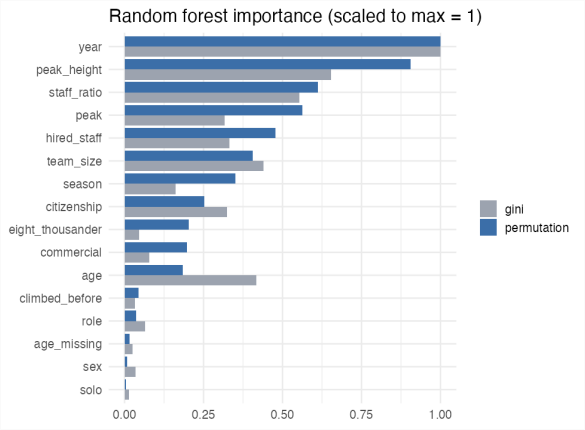
OOB would pick mtry 7 / node 25, which is worse on held-out expeditions.

Boosting (best number of trees per setting)

depth	ν	trees	val. log loss
6	**0.02**	**2250**	**0.568**
4	0.02	2750	0.569
4	0.1	850	0.569
6	0.1	300	0.570
2	0.02	3000	0.575
2	0.1	1050	0.575

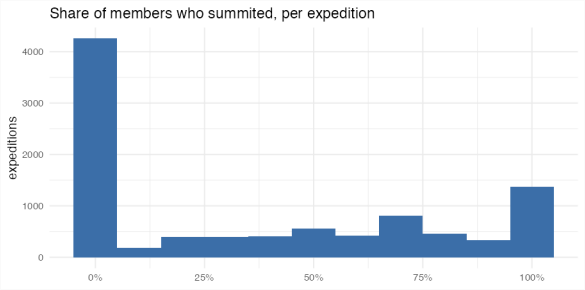
Elastic net: $\alpha \in \{0, .25, .5, .75, 1\}$; CV deviance 1.193 for all $\alpha > 0$, 1.195 for ridge.

Backup: variable importance (random forest)



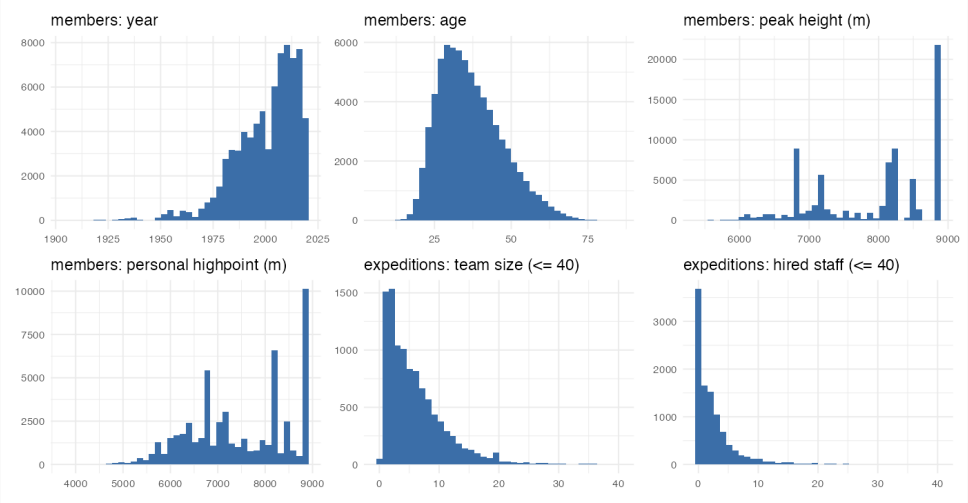
- Permutation and Gini agree on the top: year, peak height, staff per climber, peak
- Gini overrates continuous variables, e.g. age (0.42 vs 0.18), because they offer many split points
- Sex, solo and leader role barely matter

Backup: why split by expedition?

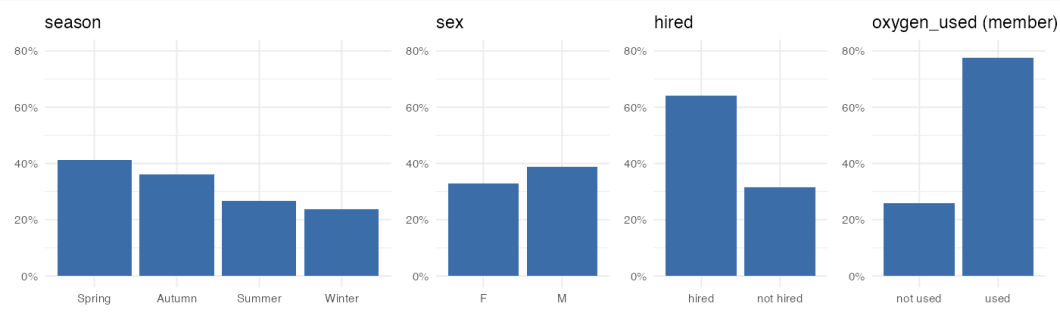


- Expeditions mostly succeed or fail together (peaks at 0% and 100%)
- A random split by climber would put teammates in training and test → over-optimistic test error
- Same reason why out-of-bag error fails (slide 7)

Backup: raw data distributions



Backup: success by category (all members, raw data)



Hired staff and climbers using oxygen succeed far more often, which is why we exclude hired staff and the oxygen variables.